Balance de cantidad movimiento angular:

$$L_i = \int_V e_{ijk} x_j \overbrace{X_k}^{r \times X} dV + \int_S e_{ijk} x_j \overbrace{\overset{n}{T}_k}^{r \times T} dS$$

$$H_i = \int_V \overbrace{e_{ijk} x_j \rho v_k}^{r \times \rho v} dV$$

$$\frac{D}{Dt} H_i = L_i$$

$$\frac{D}{Dt}\left(\int_V \rho \cdot f \, dV\right) = \int_V \rho \cdot \frac{Df}{Dt} dV$$

$$\frac{D}{Dt}(H_i) = \frac{D}{Dt}\int_V \rho \, \overbrace{e_{ijk} x_j v_k}^{f} \, dV = \int_V \rho \cdot \frac{D}{Dt}(e_{ijk} x_j v_k) dV$$

$$\frac{D}{Dt}(H_i) = \int_V \rho \cdot e_{ijk} \cdot \Big(\underbrace{\overbrace{\frac{D}{Dt} x_j}^{v_j} \cdot v_k}_{0} + x_j \cdot \frac{Dv_k}{Dt} \Big) dV$$

$$= \int_V \rho \, e_{ijk} x_j \frac{Dv_k}{Dt} dV$$

$$L_i = \int_V e_{ijk} x_j X_k \, dV + \int_S e_{ijk} x_j \overset{n}{T}_k \, dS$$

Fórmula de Cauchy: $T_i = n_j \sigma_{ji}$

$$\int_S e_{ijk} x_j n_l \sigma_{lk} \, dS \overset{\text{Teor. Gauss}}{=} \int_V \frac{\partial}{\partial x_l}(e_{ijk} x_j \sigma_{lk}) dV$$

$$= \int_V e_{ijk} \left(\overbrace{\frac{\partial x_j}{\partial x_l}}^{\delta_{jl}} \cdot \sigma_{lk} + x_j \cdot \frac{\partial \sigma_{lk}}{\partial x_l} \right) dV$$

$$= \int_V \left(e_{ijk} \sigma_{jk} + e_{ijk} x_j \cdot \frac{\partial \sigma_{lk}}{\partial x_l} \right) dV$$

$$\int_V \left[e_{ijk}\, x_j \cdot \underbrace{\left(\rho \frac{Dv_k}{Dt} - \frac{\partial \tau_{lk}}{\partial x_l} - X_k \right)}_{0} - e_{ijk}\, \tau_{jk} \right] dV = 0$$

$$-\int_V e_{ijk}\, \tau_{jk}\, dV = 0$$

$$e_{ijk}\, \tau_{jk} = 0$$

$$\begin{array}{l} j = k \\ \vee\; i = j \;\vee\; i = k \end{array} \left\{ \begin{array}{l} \overbrace{e_{ijk}}^{0}\, \tau_{jk} = 0 \qquad 0 = 0 \\ \frac{1}{2}\left(e_{ijk}\, \tau_{jk} + e_{ijk}\, \tau_{jk} \right) = 0 \end{array} \right.$$

$$\frac{1}{2}\left(e_{ijk}\, \tau_{jk} - e_{ikj}\, \tau_{jk} \right) = 0$$

$$\frac{1}{2}\left(e_{ijk}\, \tau_{jk} - e_{ijk}\, \tau_{kj} \right) = 0$$

$i \neq j$ $\quad j \neq k$

$i \neq k$

$$\underbrace{\cancel{e_{ijk}}}_{\pm 1} \left(\tau_{jk} - \tau_{kj} \right) = 0 \Rightarrow \tau_{jk} = \tau_{kj} \quad \forall\, j \neq k$$